RETRIEVAL- AUGMENTED MEDICAL QUESTION ANSWERING ASSISTANT

PROJECT PROPOSAL / DOCUMENTATION

Group-1
2420030097 – DEEPIKA.N
2420030180– K. NAVYA
2420030181 – K. BHAVYASRI

1. Project Title

Retrieval-Augmented Medical Question Answering Assistant

2. Problem Statement

Many users receive inaccurate or misleading medical information from search engines and general AI chatbots. This project proposes a Retrieval-Augmented Generation (RAG) based Medical Question Answering Assistant that provides reliable and context-aware medical answers. The system uses the MedQuAD dataset as a trusted medical knowledge base. BioBERT is used for understanding medical text, FAISS is used for retrieving relevant information, and LangChain with a Large Language Model (LLM) is used for generating context-aware responses. The application is designed using Python, Flask, HTML, CSS, and JavaScript with a simple and user-friendly interface. Additional features include user login, search history, suggested questions, download support, medical disclaimers, emergency warnings, and dark/light mode support. The proposed system aims to reduce unsupported AI responses by retrieving information from trusted medical sources before generating answers.

3. Objectives

- Develop a web-based Medical Question Answering Assistant for reliable medical information retrieval.
- Integrate the MedQuAD dataset as the primary medical knowledge base.
- Implement BioBERT embeddings to capture the semantic meaning of medical queries and documents.
- Construct a FAISS vector database for efficient semantic similarity-based retrieval.
- Orchestrate the RAG pipeline using LangChain to combine retrieved medical context with user queries.
- Generate context-aware medical responses using a Large Language Model (LLM).
- Minimize unsupported or hallucinated responses by grounding generated answers in retrieved medical information.
- Design an interactive web interface with authentication, search history, suggested questions, and responsive UI features.

3.1 Proposed Model / Framework to be Built

The proposed system follows a Retrieval-Augmented Generation architecture. Instead of directly asking an LLM to generate an answer, the system first searches a trusted medical knowledge base for relevant information. The retrieved context is then combined with the user's query and passed through the generation stage. This approach helps the response remain grounded in retrieved medical information.

4. Literature Survey

RESEARCH PAPER	YEAR	TECHNIQUES USED	PURPOSE	LIMITATION
BioBERT: A Pre-trained Biomedical Language Representation Model	2019	BioBERT, NLP	Understands biomedical text and improves medical NLP tasks.	Does not retrieve medical documents or use RAG.
Pre-trained Language Model for Biomedical Question Answering	2019	BioBERT, Fine-tuning	Answers biomedical questions using the BioASQ dataset.	No semantic retrieval or source citations.
Towards Physician-Level Medical Question Answering with LLMs(Med-PaLM)	2023	Large Language Model (LLM)	Generates medical answers with high accuracy.	Proprietary and requires high computational resources.
Improving Retrieval-Augmented Generation in Medicine (i-MedRAG)	2024	RAG, LLM	Improves medical question answering using retrieval.	Complex architecture and higher processing time.
Efficient Biomedical Question Answering using RAG	2025	RAG, FAISS, BioBERT	Retrieves relevant medical information before answering.	Research-focused; not a complete web application.
An Overview of the BIOASQ Large-Scale Biomedical Semantic Indexing and Question Answering Competition	2015	Semantic Indexing, Information Retrieval, QA	Introduces BioASQ as a benchmark for biomedical semantic indexing and question answering.	Focuses mainly on biomedical research information and does not provide a complete user-facing QA system.
Document Retrieval for Biomedical Question Answering with Neural Sentence Matching	2019	Neural Sentence Matching, Document Retrieval	Retrieves relevant biomedical documents and snippets to improve question answering.	Focuses mainly on document retrieval rather than complete answer generation.
BioASQ-QA: A Manually Curated Corpus for Biomedical Question Answering	2023	Biomedical QA, Information Retrieval	Provides a curated biomedical question-answering corpus with questions and	Mainly focuses on expert biomedical information needs and may not cover

			supporting information.	general-user medical queries.
Towards Expert-Level Medical Question Answering with Large Language Models	2023	Large Language Models, Medical Fine-tuning	Generates medical answers with high performance across multiple medical QA benchmarks.	Requires high computational resources and further validation for real-world clinical use.
Can Large Language Models Reason About Medical Questions?	2024	GPT-3.5, Llama 2, Chain-of-Thought, Retrieval Augmentatio Pre-trained Language Model for Biomedical Question Answering n	Evaluates LLM reasoning for medical questions and explores retrieval augmentation for improving QA performance.	General-purpose LLMs can still have reliability and reasoning limitations in complex medical scenarios.

4.1 Research Paper Links

1. BioBERT: A Pre-trained Biomedical Language Representation Model— 2019 —
Research Paper – Oxford Academic

2. Pre-trained Language Model for Biomedical Question Answering— 2019 —
Research Paper – arXiv

3. Towards Physician-Level Medical Question Answering with LLMs(Med-PaLM) — 2023—
Research Paper – Google Research

4. Improving Retrieval-Augmented Generation in Medicine (i-MedRAG) — 2024—
Research Paper – PubMed Central
5. Efficient Biomedical Question Answering using RAG — 2025—
Research Paper – IEEE Xplore

6. An Overview of the BIOASQ Large-Scale Biomedical Semantic Indexing and Question Answering Competition — 2015
Research Paper – PubMed Central
Research Paper – Springer / BMC Bioinformatics

7. Document Retrieval for Biomedical Question Answering with Neural Sentence Matching— 2019
Research Paper – PubMed Central
Research Paper – PubMed □

8. BioASQ-QA: A Manually Curated Corpus for Biomedical Question Answering — 2023
Research Paper – Nature Scientific Data
Research Paper – PubMed □

9. Towards Expert-Level Medical Question Answering with Large Language Models— 2023

[Research Paper – Paper / Med-PaLM 2] □

10. Can Large Language Models Reason About Medical Questions? — 2024

[Research Paper – PubMed]

[Research Paper – ScienceDirect] □

Research note: The literature table was compiled from current publisher/repository pages. Limitations are concise summaries; where a limitation was not explicit in the abstract, it is conservatively inferred from dataset scope, methodology or validation design.

5. Research Gap Identification

From the literature survey, the following research gaps were identified:

- Existing systems may generate incorrect or unsupported medical responses.
- Many models do not retrieve information from trusted medical datasets before answering.
- Most systems do not provide source references for generated answers.
- Several solutions are research-focused and not suitable for everyday users.

6. Innovation

From the literature survey, the following research gaps were identified:

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7. Creativity

The project combines modern AI technologies with a user-friendly healthcare platform. The creative aspect lies in presenting complex AI-based medical question answering through a simple interface while adding user-oriented safety and usability features.

- AI-assisted medical question answering.
- Glassmorphism-based responsive interface.
- Suggested medical questions.
- Search history.
- Source citations.
- Medical disclaimer.
- Emergency warning for critical symptoms.
- Dark/Light mode support

8. Novelty

The novelty of the proposed system lies in integrating BioBERT, FAISS, LangChain, an LLM, and a trusted medical dataset into a complete web-based medical question-answering workflow. Rather than functioning as a standalone research model, the system connects retrieval, generation, source support, and user interaction in one application. This combination is intended to improve answer accuracy, transparency, and user trust.

9. Feasibility

9.1 Technical Feasibility

- Uses open-source technologies such as Python, Flask, LangChain, BioBERT, and FAISS.

- The MedQuAD dataset is publicly available.
- The system can be developed using standard software development tools.

9.2 Economic Feasibility

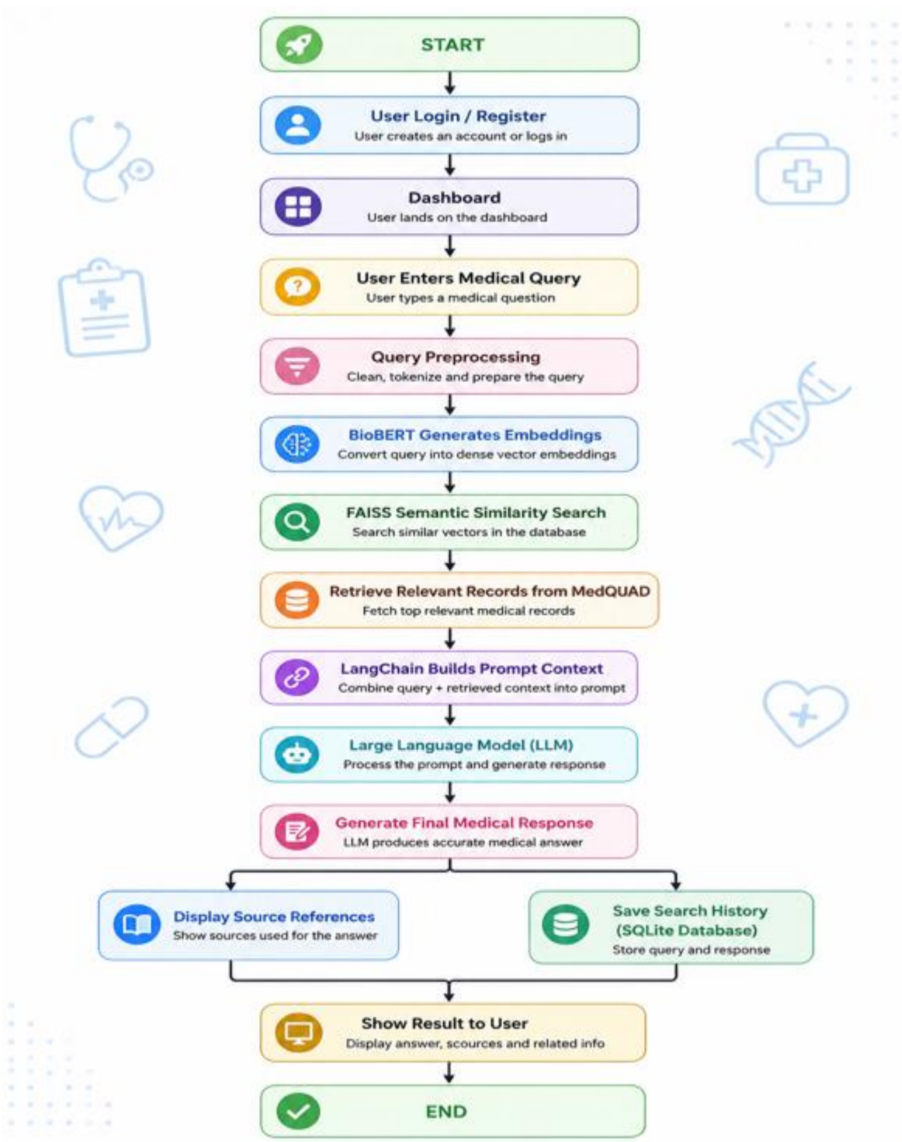
- The project uses freely available datasets and libraries.
- No expensive software or specialized hardware is required for the proposed academic prototype.

9.3 Operational Feasibility

- The system is designed around a simple web interface
- The application is suitable for students and general users.
- The prototype can run on a standard laptop subject to the computational requirements of the selected LLM.

10. Project Plan – Complete Workflow of Execution

The project will be executed as a sequence of data preparation, modelling, explanation, validation and presentation stages.



Phase 1 – Requirement Analysis: Identify the project objectives, scope, and user requirements.

Phase 2 – Literature Survey: Study existing medical question-answering approaches and related research.

Phase 3 – Dataset Collection: Collect and understand the MedQuAD dataset for the medical knowledge base.

Phase 4 – Data Preprocessing: Clean, organize, and prepare the dataset for embedding generation.

Phase 5 – BioBERT Embedding Generation: Convert medical text into semantic vector representations using BioBERT.

Phase 6 – FAISS Vector Database: Store embeddings and enable fast semantic similarity retrieval.

Phase 7 – RAG Pipeline Development: Integrate LangChain and the LLM to create the retrieval-augmented workflow.

Phase 8 – Frontend Development: Develop a responsive interface using HTML, CSS, and JavaScript.

Phase 9 – Backend Development: Develop the Flask backend and connect the application modules.

Phase 10 – Integration and Testing: Integrate AI, frontend, backend, and supporting features and test the system.

Phase 11 – Documentation: Prepare the project documentation and presentation.

11. Expected Deliverables

- Processed medical knowledge base based on the MedQuAD dataset.
- BioBERT-based semantic representations.
- FAISS vector database for medical information retrieval.
- LangChain-based RAG pipeline.
- LLM-based medical response generation module.
- Flask backend and web interface.
- User authentication and search-history functionality.
- Suggested questions, source citations, medical disclaimer, and emergency warning features.
- Project documentation and presentation.

12. Expected Outcomes

- A web-based Medical Question Answering Assistant capable of retrieving relevant medical information before generating responses.
- More context-aware answers grounded in retrieved medical information.
- Reduced dependence on unsupported direct generation by the LLM.
- A simple and user-friendly interface for medical information queries.
- Improved transparency through source-supported responses and medical disclaimers.

13. Key Limitations of the Proposed Project

- **Dataset Limitation:** The system depends on the MedQuAD dataset, so the quality and coverage of answers are limited by the information available in the dataset.
- **Medical Knowledge Limitation:** The system may not cover every medical condition, treatment, symptom, or newly emerging medical information.
- **LLM Limitation:** Although the RAG approach reduces unsupported responses, the Large Language Model may still generate inaccurate or incomplete information.
- **Retrieval Limitation:** If the relevant information is not retrieved from the knowledge base, the generated response may not provide the required answer accurately.
- **Source Limitation:** The reliability of the generated answer depends on the quality and reliability of the medical information used as the knowledge source.
- **No Medical Diagnosis:** The system is intended to provide medical information and should not be considered a replacement for professional medical diagnosis or treatment.
- **Real-Time Information Limitation:** The proposed system may not contain the latest medical research, guidelines, or newly discovered treatments unless the knowledge base is updated.
- **Privacy and Security:** If user accounts and search history are stored, appropriate privacy and security mechanisms are required to protect user information.
- **Clinical Validation:** The proposed academic prototype requires extensive testing and validation before it could be considered for real-world clinical use.

14. Conclusion

The Retrieval-Augmented Medical Question Answering Assistant is proposed as a reliable and context-aware approach to medical information retrieval and question answering. By combining the MedQuAD dataset, BioBERT, FAISS, LangChain, and a Large Language Model, the system retrieves relevant medical information before generating an answer. The addition of authentication, search history, suggested questions, source citations, medical disclaimers, emergency warnings, and responsive UI features makes the proposed solution more practical and user-oriented. The system is intended as an academic prototype for providing grounded medical information and should not be treated as a replacement for professional medical diagnosis or advice.

15. Key Dataset Resources

- MedQuAD (Medical Question Answering Dataset)- abachaa/MedQuAD: Medical Question Answering Dataset of 47,457 QA pairs created from 12 NIH websites
- PubMed-PubMed